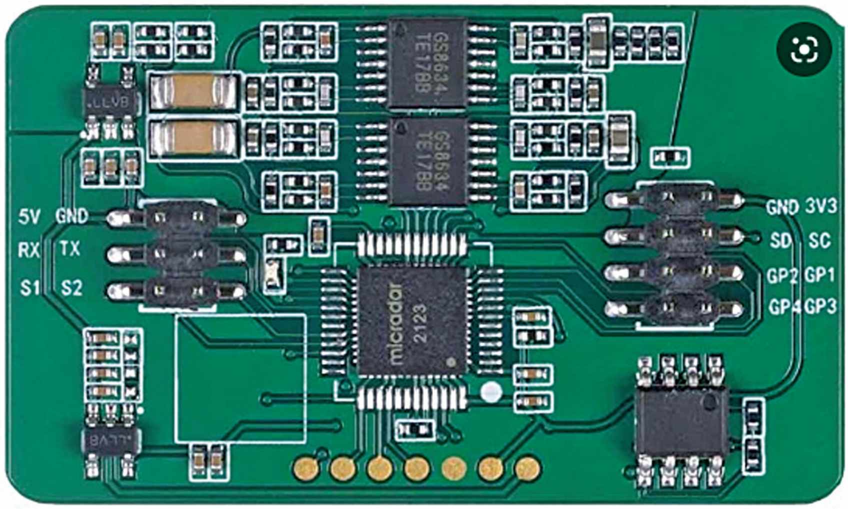

Fig. 5: mm-wave radar board (Image credit: seedstudio.com)

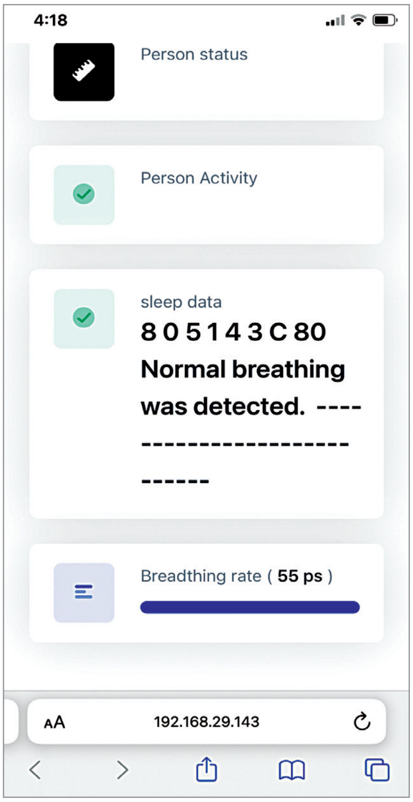

Fig. 6: Dashboard of the data activity

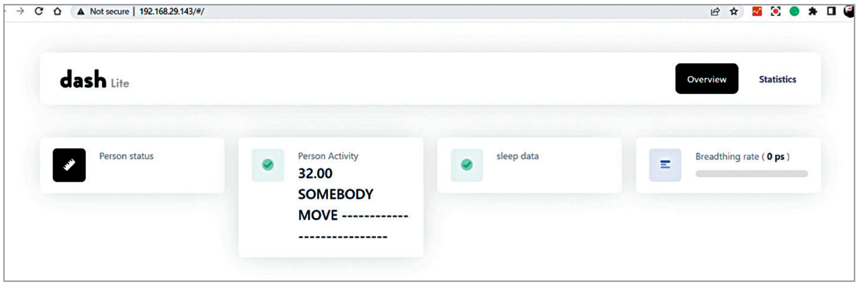

Fig. 7: Dashboard of the moving activity

TX2 pins of ESP board, so that you would be able to get the radar readings through the serial port (see Fig. 2).

So, first open the library folder and then open the library main ccp file, and under that do the setting and add the code shown in Fig. 3. After modifying the library code save it.

First, you need to add the library in the code and then add the cards to show the data. Then in the loop function, check the radar data and implement the algorithm that finds the change in data and senses any human movement and action, and gives you the output. After that display and update the cards for each data in the UI of ESP das webserver that you made.

You have one card for action, which shows the actions like moving. sleeping, reading, going away, entering, dancing, and falling. Then you have another card that tells you about sleep data and respiration rate, which indicates whether the person is sitting on the bed, leaving the bed, or sleeping on the bed. The third card shows the respiration rate and the vitals detected like pulse and respiration.

As we are using 24Hz radar, it tells only the respiration rate. By using 6Hz radar we can also detect heartbeat and pulse signals along with respiration rate.

Working and testing

Connect the components as per the circuit diagram shown in Fig. 4. Connect the ESP development board's RX2 and TX2 terminals to TX and RX serial pins of the mm-wave radar board (Fig. 5). Connect RX0 and TX0 to each other in ESP32 board.

For tesing, power the device through USB of laptop and wait till the device gets connected to Wi-Fi network that you set in code. It will give you the IP address of the device. Note this IP address.

Now you can set up the device anywhere you want to track the person's sleep, activity, respiration rate, etc. You can keep it in the bathroom, bedroom, or anywhere else where you want to track without affecting the person's privacy.

After proper testing, power the device with a regular 5V DC adaptor and then search for the IP address in your phone connected to the same Wi-Fi network. Now you can monitor the person's activity on a dashboard as shown in Fig. 6 and Fig. 7. EFY

EFY Note
The source code of this project is available for free download at *www.electronicsforu.com*

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